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Comparison of Brain Structure Volumes in Insectivora and Primates. IV. Non-cortical visual structures

By

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With 7 Figures and 2 Tables

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Abstract and Summary: The relative size of the eyes, optic nerves, chiasms and tracts, and of the dorsal nucleus of the lateral geniculate body is distinctly larger in Primates than in (theoretically) isoponderous Insectivora. Within Insectivora, the relative size is lowest in moles, medium in shrews and hedgehog-like tenrecs, and largest in hedgehogs. Within Primates, all relative sizes are on the average larger in simians than in prosimians: the eyes to a small degree, the lateral geniculate bodies moderately and the optic nerves considerably larger. The ratio between eyes and optic nerves is large in night-active primates and distinctly smaller in day-active forms, with no overlap. The only night-active simian (*Aotus trivirgatus*) is in line with night-active prosimians. The relative size of the non-cortical visual structures in man is in line with that of day-active simians, whereas two of the great apes (orang-utan and gorilla) are relatively low. The size of the visual structures appears to depend mainly on functional requirements and is not, or is distinctly less, related to differences in the evolutionary level.

The size of the visual structures of tree-shrews (Scandentia) shows special features which are not found in Insectivora and Primates and is compatible with their separation from these orders.

Introduction

The visual system seems to play a minor role in Insectivora. In this order, the olfactory system appears to be the main sensory system (a comparison of olfactory bulb size is given in Part III of this series; BARON et al., 1983). In higher primates, the olfactory structures are distinctly reduced and in all primates the visual system becomes more dominant. In accordance with these differences, we were able to delineate, using methods of cytoarchitecture, a primary visual cortex in all primates, but not in Insectivora (see Part V; FRAHM et al., 1984). This cortical field seems to exist in Insectivora but it is not possible to delineate it with sufficient precision for comparative investigation. Such a delineation, however, was possible for some non-cortical visual structures, and from the relation between non-cortical and cortical visual structures in primates, some indications as to the size of the cortical visual structures in Insectivora are expected (Part V).

Further aims of the present study were: (1) to gain some insight into species and/or family differences in non-cortical visual structures within Insectivora and Primates and between these orders; and (2) to obtain data that would serveas a basis for a discussion on structural size in relation to ecology, behaviour and phylogeny.

Material and Methods

Half the surface of the eye (HSE) was calculated from the average transverse diameters by using the equation of the surface of a sphere HSE $= 1/2\pi d^2$. Diameters of the eyes were measured in 78 eyes from 23 species of Insectivora, Scandentia (tree shrews), Primates, and some laboratory animals. In addition, diameters and/or volumes of eyes from 56 species were taken from the literature. For 29 of the total of 63 species the data have come from either two or more sources (see Table 1, column 2, and footnote to Table 1). Eye volumes given in the sources were converted into diameters and the diameters collected were averaged. Because of their different origins, the data are heterogeneous and may not in all cases be well founded. Values which were obviously incorrect and/or strongly deviant from those of related species were not included in the data collection.

Eye volumes (EYV) were also calculated from the average diameters by using the volume equation of a sphere EYV $= 1/6\pi d^3$. They are not given in Table 1 but can be easily

Table 1. Data on half surface of eyes (HSE), cross sectional areas of nervus opticus (NOA), mid-sagittal sectional areas of chiasma opticum (CH II), volumes of tractus opticus (TRO), and volumes of dorsal nucleus of corpus geniculatum laterale (CGL). Area and volume data are from both sides of an individual (except CH II), those in brackets are taken from related species (also listed) for calculation of ratios.

	Eye			Nervus opticus			Chiasma opticum		Tractus opticus	Corpus gen. lat.	
	HSE mm^2 (1)	origin of data (2)	n (3)	NOA mm^2 (4)	origin of data (5)	n (6)	CH II mm^2 (7)	n (8)	TRO mm^3 (9)	CGL mm^3 (10)	n (11)
Insectivora											
0577 Tenrec ecaudatus	—	—	—	0.22	S	6	0.22	2	2.73	1.46	1
0579 Setifer setosus	—	—	—	0.09	S	4	0.08	2	1.07	1.05	1
0581 Hemicentetes semispin.	—	—	—	0.08	S	2	0.07	1	0.51	0.37	1
0585 Echinops telfairi	—	—	—	0.10	S	3	0.11	2	0.61	0.37	1
0689 Erinaceus algirus	151	S	3	0.52	S	9	0.30	2	—	5.02	1
0695 Erinaceus europaeus	99.2	1, 9, S	4*	0.34	S	8	0.23	1	3.94	3.90	1
0701 Hemiechinus auritus	113	S	2	0.34	S	7	0.20	3	—	—	—
0715 Sorex araneus	4.9	9	1*	0.02	S	7	0.03	3	0.14	0.07	1
0769 Sorex minutus	5.0	9	1*	0.03	S	2	0.03	1	0.11	0.06	1
0839 Neomys fodiens	—	—	—	0.02	S	3	0.03	1	—	—	—
0943 Crocidura flavescens	—	—	—	0.02	S	4	0.04	5	0.25	0.28	1
0952 Crocidura giffardi	7.1	19	1	—	—	—	—	—	—	—	—
1071 Crocidura russula	3.5	4 (C. leuc.)	1*	0.01	S	2	0.02	1	0.12	0.08	1
1085 Crocidura suaveolens	2.5	5	1*	—	—	—	—	—	—	—	—
1137 Suncus murinus	—	—	—	0.04	11, S	3*	0.05	1	0.23	0.18	1
1205 Desmana moschata	—	—	—	0.05	S	2	0.06	1	—	—	—
1207 Galemys pyrenaicus	—	—	—	0.03	S	2	0.03	1	—	—	—
1215 Talpa europaea	2.5	1, 9	2*	0.01	S	2	0.01	1	—	—	—
Scandentia											
1263 Tupaia glis	219	S	8	3.15	S	3	4.57	3	39.3	12.9$^+$	4
1269 Tupaia minor	—	—	—	—	—	—	—	—	—	9.88	2
1289 Urogale everetti	226	10, S	3*	3.04	10, S	5*	5.83	1	—	9.98	1
Primates											
Prosimians											
3199 Microcebus murinus	299	18, S	3*	0.68	S	12	0.94	1	6.17	7.07$^+$	5
3203 Cheirogaleus major	445	S	1	0.90	S	5	1.27	2	18.0	21.5	2
3205 Cheirogaleus medius	387	S	1	0.79	S	5	0.66	1	—	10.8	2
3209 Phaner furcifer	515	18	1*	—	—	—	—	—	—	—	—
3211 Lemur catta	757	7, 13	2*	3.70	S	2	4.79	1	—	—	—
3215 Lemur albifrons	(757)	—	—	3.14	S	6	3.96	5	108	51.8$^+$	2
3227 Varecia variegata	942	13, S	3*	3.87	S	4	3.16	2	—	68.9	1
3239 Lepilemur ruficaudatus	—	—	—	1.12	S	6	1.37	1	20.7	15.6	3
3243 Avahi l. laniger	—	—	—	—	—	—	—	—	—	27.6$^+$	1
3244 Avahi l. occidentalis	—	—	—	2.11	S	4	1.43	1	38.3	29.3$^+$	2
3247 Propithecus verreauxi	—	—	—	3.18	S	4	3.18	1	125	64.5$^+$	2
3249 Indri indri	—	—	—	3.79	S	4	2.62	1	—	94.5$^+$	2
3251 Daubentonia madagascar.	966	7	1*	2.21	S	2	1.40	1	—	58.5	1
3253 Loris tardigradus	788	3, 14, S	4*	0.94	S	4	1.06	3	15.7	24.6	1
3255 Nycticebus coucang	714	7, 13, 14, S	7*	1.46	S	6	1.63	3	—	35.9$^+$	1
3259 Perodicticus potto	531	7, 14	2*	0.93	S	4	1.20	1	19.5	24.3	2
3265 Galago senegalensis	423	7	1*	1.68	S	2	1.46	3	—	16.0	1
3267 Otolemur crassicaudatus	731	3, 13, 14, S	9*	1.70	13, S	13*	1.89	2	33.9	23.2$^+$	1
3275 Galagoides demidoff	475	13	1*	1.13	13, S	5*	1.07	2	11.5	10.9	2
3282 Tarsius spp.	990	7, 9, 10, 14, S	11*	3.30	10, S	12*	2.35	2	15.5	20.8	2
Simians											
3287 Callithrix jacchus	465	1, S	8*	3.35	S	12	3.01	3	—	25.1$^+$	2
3289 Cebuella pygmaea	—	—	—	—	—	—	—	—	23.8	16.9	1
3306 Saguinus midas tamarin	—	—	—	4.40	S	4	3.64	1	—	36.0$^+$	1

Table 1 (Continued)

	Eye			Nervus opticus			Chiasma opticum		Tractus opticus	Corpus gen. lat.	
	HSE mm² (1)	origin of data (2)	n (3)	NOA mm² (4)	origin of data (5)	n (6)	CH II mm² (7)	n (8)	TRO mm³ (9)	CGL mm³ (10)	n (11)
3311 Saguinus oedipus	452	13	2*	4.72	S	8	3.65	1	50.5	34.2+	1
3312 Saguinus oed. geoffroyi	484	7	2*	—	—	—	—	—	—	—	—
3324 Cebus spp.	1197	13, S	3*	10.9	13, S	9*	8.34	1	207	137	2
3325 Aotus trivirgatus	1193	7, 13, S	8*	3.53	S	10	3.58	1	—	32.9+	2
3327 Callicebus moloch	—	—	—	—	—	—	—	—	—	54.2+	1
3333 Saimiri oerstedi	586	7	1*	—	—	—	—	—	—	—	—
3335 Saimiri sciureus	698	13, S	11*	6.31	13, S	11*	6.11	2	117	62.9	1
3341 Pithecia monachus	—	—	—	—	—	—	—	—	—	76.3+	1
3366 Alouatta spp.	962	7, S	4*	7.65	S	6	6.21	3	—	87.4+	1
3371 Ateles geoffroyi	1143	7	1*	9.60	S	2	8.63	2	204	151	1
3379 Lagothrix lagothricha	1186	S	6	8.87	S	9	9.21	1	203	164+	1
3387 Macaca fascicularis	985	7, 13	22*	8.36	S	1	9.80	3	—	—	—
3391 Macaca mulatta	1283	7, 10, 13, S	12*	9.47	13, S	8*	7.36	2	—	158	1
3393 Macaca nemestrina	1293	7	4*	—	—	—	—	—	308	—	—
3407 Cercocebus albigena	(1187)	—	—	13.4	S	2	11.9	1	—	182	1
3414 Cercocebus spp.	1187	13	10*	18.4	13	10*	—	—	—	—	—
3415 Papio anubis	1527	13	6*	27.0	13, S	10*	14.7	1	601	395	1
3421 Papio papio	1265	13	3*	26.4	13	2*	—	—	—	—	—
3427 Mandrillus sphinx	1660	7	1*	—	—	—	—	—	—	—	—
3431 Cercopithecus aethiops	1143	7, 13	4*	14.1	13	3*	—	—	—	—	—
3433 Cercopithecus ascanius	(1007)	—	—	13.9	S	2	12.0	2	320	147	1
3453 Cercopithecus mitis	1217	13	3*	13.0	13, S	3*	10.2	1	—	150	1
3459 Cercopithecus nictitans	1007	13	9*	12.3	13	9*	—	—	—	—	—
3468 Cercopithecus spec.	1196	7	1*	—	—	—	—	—	—	—	—
3469 Miopithecus talapoin	840	S	4	6.29	S	7	5.71	1	—	109+	1
3473 Erythrocebus patas	1605	13, S	5*	13.7	13, S	8*	14.7	1	—	—	—
3477 Colobus badius	—	—	—	10.1	S	4	11.4	1	201	128	2
3499 Nasalis larvatus	1120	7	23*	12.1	S	2	10.6	1	304	—	—
3501 Presbytis aygula	1261	7	1*	—	—	—	—	—	—	—	—
3503 Presbytis cristata	1161	7	19*	—	—	—	—	—	—	—	—
3527 Presbytis rubicundus	1374	7	11*	—	—	—	—	—	—	—	—
3539 Hylobates lar	1299	7	32*	12.7	S	2	6.94	1	302	175	1
3540 Hylobates moloch	1259	7	5*	—	—	—	—	—	—	—	—
3544 Hylobates spec.	1119	13	2*	17.1	13	1*	—	—	—	—	—
3545 Pongo pygmaeus	1282	7, 13	7*	16.1	13	2*	—	—	—	—	—
3549 Pan troglodytes	1446	7, 10, 13	16*	21.0	13, S	8*	12.6	3	630	356	1
3551 Gorilla gorilla	1774	10, 13	4*	17.6	10, 13, S	4*	13.1	3	671	384	1
3553 Homo sapiens	1855	1, 6, 7, 9, 10, 20	14*	22.8	10, 20, S	5*	8.63	3	823	416	1
Domestic and laboratory animals											
3562 Canis familiaris	1543	8, 9	5*	8.62	S	1	9.92	1	—	—	—
4010 Felis domestica	1359	2, 9, S	4*	4.51	16, 17, S	8*	4.00	2	—	—	—
6918 Rattus norvegicus alba	141	2, 12	11*	0.25	12, S	10*	0.52	2	—	—	—
7244 Mus musculus alba	34.8	9, 12	6*	0.11	12	5*	—	—	—	—	—
7984 Oryctolagus cuniculus	855	9	1*	1.72	15, S	3*	1.96	1	—	—	—
Macroscelidea											
7993 Elephantulus fuscipes	—	—	—	0.64	S	4	1.17	1	—	—	—
8011 Rhynchocyon cirnei	—	—	—	3.90	S	4	5.59	1	—	—	—

* Numbers given are minimums since it is not always clear from data taken from the literature how many eyes or optic nerves were measured. Key to the source of the raw data of eyes and optic nerves: 1 Franz (1911, 1934); 2 Kahmann (1930); 3 Kolmer (1930); 4 Schwarz (1935); 5 Verrier (1935); 6 Lauber (1936); 7 Schultz (1940); 8 Arey et al. (1942); 9 Rochon-Duvigneaud (1943); 10 Polyak (1957); 11 Sharma (1958); 12 Jansky (1959); 13 Rohen (1962); 14 Spatz (1968, 1970); 15 Hughes and Wässle (1976); 16 Vaney and Hughes (1976); 17 Stone and Campion (1978); 18 Pariente (1979); 19 Waterlot (1912); 20 Salzmann (1912); S Stephan and collaborators.

\+ These CGL data deviate from those given in the data collection of Stephan et al. (1981) due to widening of the material and/or re-investigation with more highly enlarged photographic series.

converted from the given HSE values by using EYV $= \frac{HSE^{3/2}}{5.317}$. This conversion accounts for the two eyes of a given individual (see later). 5.317 is $1.5\sqrt{4\pi}$ and must be substituted for the conversion within a single eye by $1.5\sqrt{2\pi}$, which is 3.760.

The *cross sectional area of the nervus opticus (NOA)* was calculated from its diameter by using the formula for the surface of a circle $NOA = 1/4\pi d^2$. The diameters of 68 nerves were measured close to the eyeball in 23 species. Further diameters and/or transectional areas of 222 nerves were measured in serial sections of 64 species. Both linear and areal measurements were converted to fresh brain size. Conversion factors were calculated from the volume shrinkage of the individual brain. In addition, diameters and/or areas of optic nerves from 25 species were taken from the literature (mainly from ROHEN, 1962). For 20 of the total of 72 species, data were taken from either two or more sources (see Table 1, column 5).

The *chiasma opticum (CH II)* was measured in the mid-sagittal plane of serial-sectioned brains of 87 animals representing 50 species. In 25 individuals of 17 species, data were obtained from reconstructions of frontally sectioned brains. All measurements were converted to fresh brain size.

Volumes of the *optic tract (TRO)* from 37 species were determined by one of the authors (BARON) when subdividing the diencephalon. These figures were published in STEPHAN et al., 1981. Details on the material are given there.

Volumes of the *dorsal nucleus of the corpus geniculatum laterale (CGL)* from 78 individuals of 55 species were determined from serial sections of frontally-sectioned brains. Deviations from the data published in the collection of STEPHAN et al. (1981) are due to widening of the material and/or re-investigation in more highly enlarged photographic series. All measured volumes were converted to fresh brains.

The data given in Table 1 for HSE, NOA, TRO and CGL are averages each for the whole individual and not for one side only since brain structures from the mid-sagittal plane could not be separated with sufficient precicion to incorporate them in one or the other side. Therefore, all the brain data given so far are from the whole brain. This is also important since these data are plotted with the body weight, i.e. the weight of the total body and not of its half. The CH II areas were taken from the mid-sagittal plane. They contain fibers of the optic nerves of both sides and are, of course, not doubled.

Data from laboratory animals are added at the end of the Tables. They will be used e.g. in the discussion on fiber numbers, since fiber contents of optic nerves are mainly investigated in laboratory animals.

Complete sets of data were available only from a few species, but the data on hand seem to reflect general trends and make it possible to evaluate special qualities of single species or groups of species.

For the size *comparisons*, three different methods are used: (1) direct comparison of two different structures by *ratios*; (2) *interstructural regression* line analyses between two structures in double logarithmic diagrams; and (3) *size indices* (SI), which evaluate size differences of the given structures in isoponderous species. The methods are described in detail in Part I and II (FRAHM et al., 1982; STEPHAN et al., 1982). In these studies, Insectivora were used as the reference group. Since exact measurements of the primary visual cortex could not be made in Insectivora,

Table 2. Size indices of half the surface of the eyes (HSE), cross sectional areas of nervus opticus (NOA), and volumes of the dorsal nucleus of the corpus geniculatum laterale (CGL), and ratios HSE/NOA and CH II/NOA. The ratios are calculated directly from the data given in Table 1. The indices reflect the distances from the reference lines in Figures 1–3 and are in percent, i.e. all points on the lines have values of 100.

		Indices			Ratios	
		HSE	NOA	CGL	HSE/NOA	CH II/NOA
		(1)	(2)	(3)	(4)	(5)
Insectivora						
0577	Tenrec ecaudatus	—	11.8	4.9	—	1.01
0579	Setifer setosus	—	6.6	6.3	—	0.91
0581	Hemicentetes semispin.	—	8.3	3.2	—	0.87
0585	Echinops telfairi	—	10.5	3.7	—	1.09
0689	Erinaceus algirus	22.5	28.9	17.2	289	0.58
0695	Erinaceus europaeus	14.5	18.0	12.8	296	0.68
0701	Hemiechinus auritus	21.2	26.5	—	332	0.60
0715	Sorex araneus	1.7	4.1	2.0	246	1.35
0769	Sorex minutus	2.0	7.2	2.2	172	0.93
0839	Neomys fodiens	—	4.0	—	—	1.18
0943	Crocidura flavescens	—	3.3	4.7	—	1.82
0952	Crocidura giffardi	1.7	—	—	—	—
1071	Crocidura russula	1.5	2.4	2.1	294	1.50
1085	Crocidura suaveolens	1.1	—	—	—	—
1137	Suncus murinus	—	5.2	2.7	—	1.22
1205	Desmana moschata	—	3.6	—	—	1.00
1207	Galemys pyrenaicus	—	3.8	—	—	0.97
1215	Talpa europaea	0.6	1.2	—	231	0.55
Scandentia						
1263	Tupaia glis	44.3	275	92.4	69.6	1.45
1269	Tupaia minor	—	—	108	—	—
1289	Urogale everetti	41.5	230	56.8	74.5	1.92
Primates						
Prosimians						
3199	Microcebus murinus	76.1	84.2	87.9	438	1.38
3203	Cheirogaleus major	74.1	58.7	96.5	496	1.42
3205	Cheirogaleus medius	77.7	67.8	75.9	493	0.84
3209	Phaner furcifer	85.8	—	—	—	—
3211	Lemur catta	95.5	159	—	205	1.29
3215	Lemur albifrons	—	146	135	241	1.26
3227	Varecia variegata	107	143	125	243	0.82
3239	Lepilemur ruficaudatus	—	59.1	49.8	—	1.22
3243	Avahi l. laniger	—	—	74.9	—	—
3244	Avahi l. occidentalis	—	113	96.3	—	0.68
3247	Propithecus verreauxi	—	112	109	—	1.00
3249	Indri indri	—	112	120	—	0.69
3251	Daubentonia madagascar.	112	83.3	109	438	0.63

Table 2 (Continued).

		Indices			Ratios	
		HSE	NOA	CGL	HSE/NOA	CH II/NOA
		(1)	(2)	(3)	(4)	(5)
3253	Loris tardigradus	140	67.6	130	842	1.13
3255	Nycticebus coucang	106	80.4	122	488	1.11
3259	Perodicticus potto	73.3	46.0	69.5	569	1.28
3265	Galago senegalensis	84.1	143	110	252	0.87
3267	Otolemur crassicaudatus	107	91.9	76.9	429	1.11
3275	Galagoides demidoff	111	124	111	420	0.94
3282	Tarsius spp.	213	316	173	300	0.71
	Simians					
3287	Callithrix jacchus	85.1	252	141	139	0.90
3289	Cebuella pygmaea	—	—	133	—	—
3306	Saguinus midas tamarin	—	312	185	—	0.83
3311	Saguinus oedipus	77.9	324	166	95.7	0.77
3312	Saguinus oedipus geoffroyi	74.9	—	—	—	—
3324	Cebus spp.	135	398	244	110	0.77
3325	Aotus trivirgatus	176	192	110	338	1.01
3327	Callicebus moloch	—	—	—	—	—
3333	Saimiri oerstedi	85.1	—	—	—	—
3335	Saimiri sciureus	108	367	235	111	0.97
3341	Pithecia monachus	—	—	—	—	—
3366	Alouatta spp.	94.2	225	110	126	0.81
3371	Ateles geoffroyi	107	264	170	119	0.90
3379	Lagothrix lagothricha	121	278	228	134	1.04
3387	Macaca fascicularis	108	291	—	118	1.17
3391	Macaca mulatta	121	263	181	135	0.78
3393	Macaca nemestrina	121	—	—	—	—
3407	Cercocebus albigena	—	371	207	88.5	0.88
3414	Cercocebus spp.	111	508	—	64.6	—
3415	Papio anubis	114	528	258	56.5	0.55
3421	Papio papio	101	570	—	47.9	—
3427	Mandrillus sphinx	118	—	—	—	—
3431	Cercopithecus aethiops	124	485	—	80.9	—
3433	Cercopithecus ascanius	—	494	250	72.5	0.86
3453	Cercopithecus mitis	120	383	190	93.9	0.79
3459	Cercopithecus nictitans	104	390	—	81.8	—
3468	Cercopithecus spec.	125	—	—	—	—
3469	Miopithecus talapoin	115	306	307	134	0.91
3473	Erythrocebus patas	151	379	—	118	1.08
3477	Colobus badius	—	290	154	—	1.12
3499	Nasalis larvatus	92.3	275	—	92.7	0.88
3501	Presbytis aygula	123	—	—	—	—
3503	Presbytis cristata	114	—	—	—	—
3527	Presbytis rubicundus	137	—	—	—	—
3539	Hylobates lar	130	386	233	102	0.55
3540	Hylobates moloch	126	—	—	—	—

Table 2 (Continued).

		Indices			Ratios	
		HSE	NOA	CGL	HSE/NOA	CH II/NOA
		(1)	(2)	(3)	(4)	(5)
3544	Hylobates spec.	112	519	—	65.4	—
3545	Pongo pygmaeus	80.8	244	—	79.7	—
3549	Pan troglodytes	95.4	343	173	68.7	0.60
3551	Gorilla gorilla	99.2	224	126	101	0.75
3553	Homo sapiens	114	335	172	81.3	0.38
	Domestic and laboratory animals					
3562	Canis familiaris	120	180	—	179	1.15
4010	Felis domestica	146	153	—	301	0.89
6918	Rattus norvegicus alba	25.6	18.6	—	564	2.06
7244	Mus musculus alba	11.0	18.0	—	331	—
7984	Oryctolagus cuniculus	101	67.0	—	498	1.14
	Macroscelidea					
7993	Elephantulus	—	77.6	—	—	1.83
8011	Rhynchocyon cirnei	—	248	—	—	1.43

the prosimians were used as a reference group for the comparison of *all* the visual structures (Parts IV and V).

The vertical distance of each species from the prosimian reference line indicates the size difference of the given structure relative to that of an "average" prosimian with equivalent body weight. These distances are expressed in the form of size indices (SI).

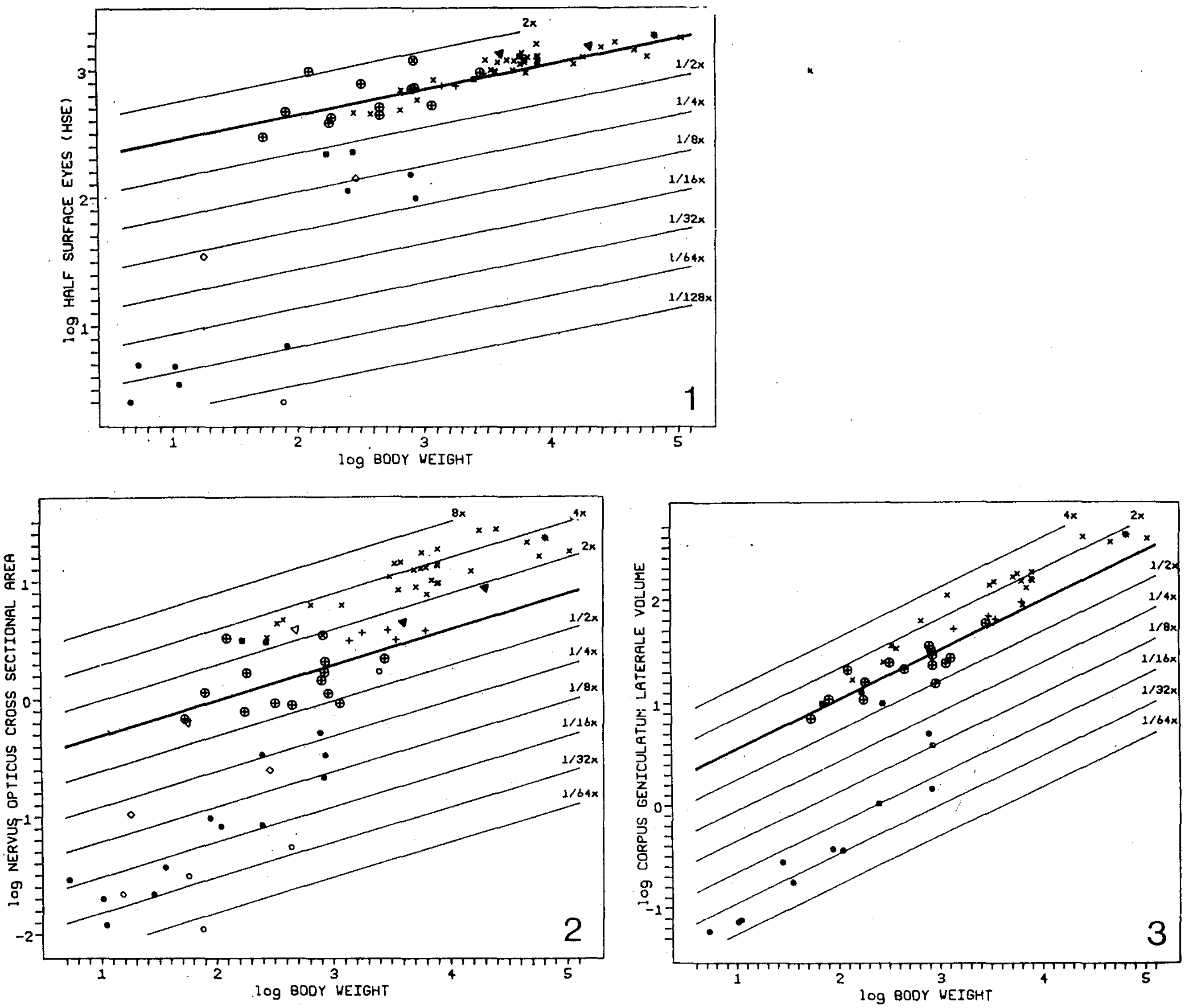
Results

Eyes and optic nerves

Instead of eye volumes, half the surface of the eye was compared (Tables 1 and 2, columns 1) to provide a rough comparative measure of the extent of the retina.

Size indices of half the surface of the eyes (HSE): The slope of the regression line of all prosimian species (n 14) was 0.189, and that of all simian species (n 34) was 0.211. The slopes of the regression lines of the seven primate families under investigation (with more than one species) had an average of 0.192. The average of the values is close to 0.20 and, therefore, the reference line was given this slope. A line with this slope was drawn through the geometric means of body weights and HSEs of the prosimians (Fig. 1). Thus for each body weight this line indicates the HSE of a typical prosimian.

The size indices related to the prosimian base line (A = 0.20; B = 2.248) are given in Table 2, column 1, and in Fig. 4. The average index of prosimians was 96 without *Tarsius* and 104 with this genus, while that of simians was 112. The index



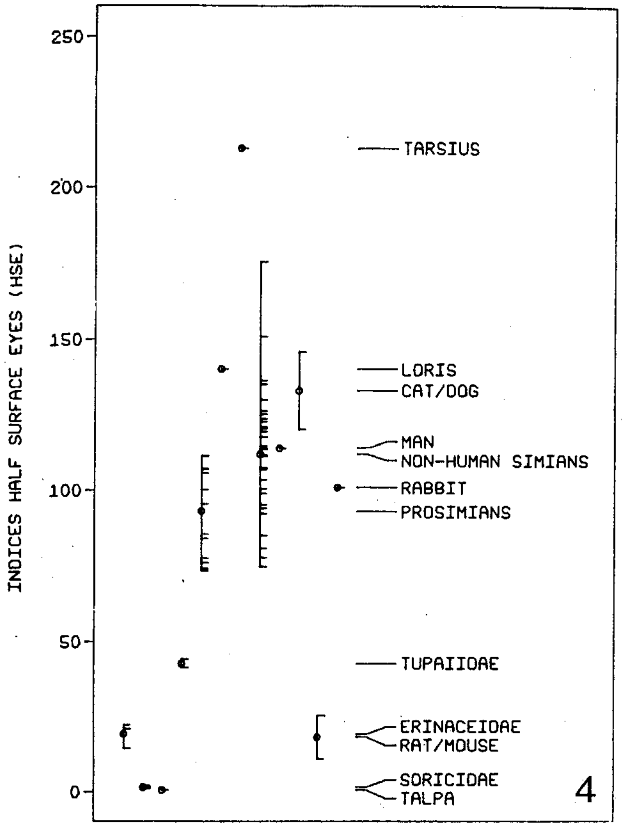
Figs. 1–3. Half surface of the eye (HSE, Fig. 1), nervus opticus cross sectional area (NOA, Fig. 2), and corpus geniculatum laterale volume (CGL, Fig. 3) plotted against body weight (BoW) in double logarithmic scales. The reference line in each plot (thick line) is through the average structure size and average body weight of the prosimians with the common slope found for the primate families and suborders (see text). The thinner parallel lines are in multiples of 2 and refer to the size indices (×100) given in Table 2 and scaled in Figures 4–6.

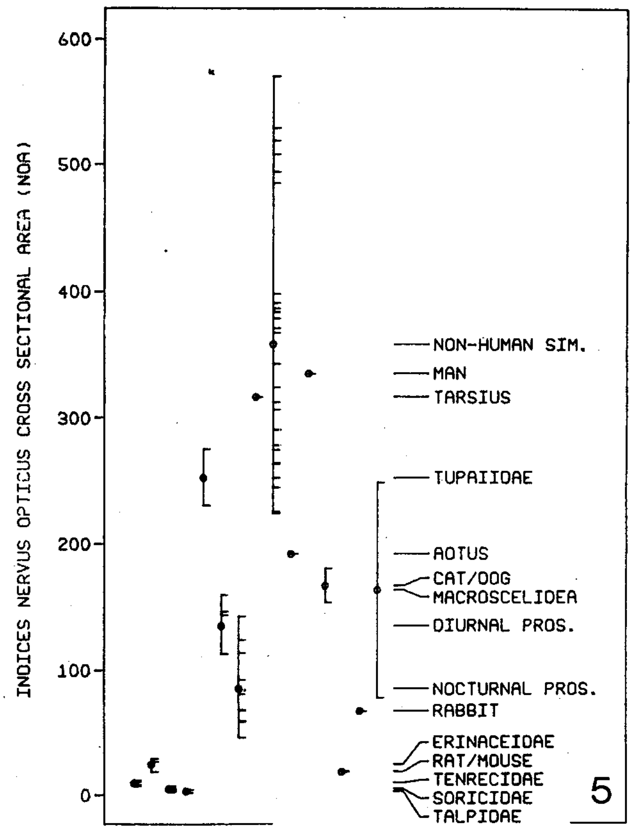
● basal Insectivora (Tenrecinae, Soricidae, Erinaceidae); ○ progressive Insectivora (*Neomys*, Talpidae); ■ tree-shrews; + prosimians; × non-human simians; * man; ◄ dog and cat; ◇ mouse and rat; □ rabbit; ◁ elephant shrews. Nocturnal primates are encircled.

Formulas of the reference lines are

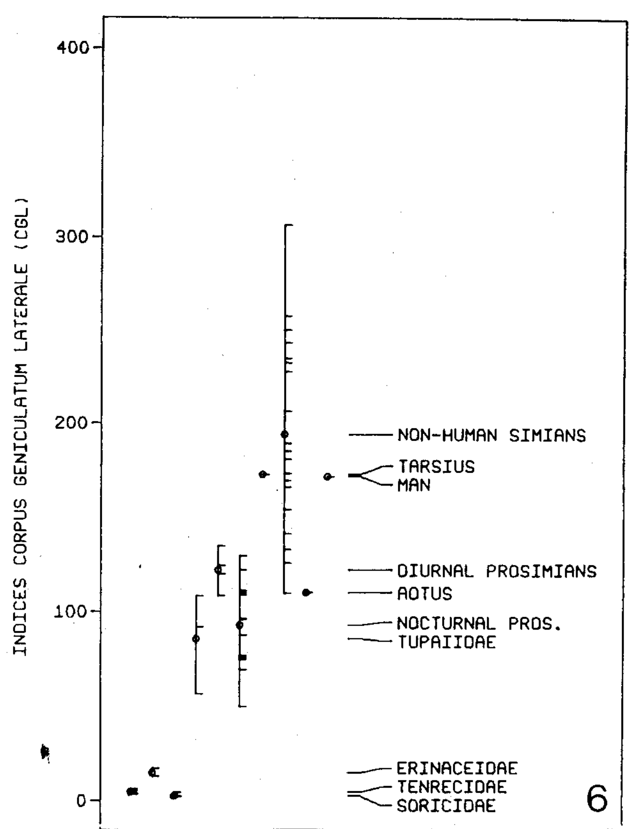
Fig. 1 $\log \text{HSE} = 2.248 + 0.20 \cdot \log \text{BoW}$
Fig. 2 $\log \text{NOA} = -0.611 + 0.30 \cdot \log \text{BoW}$
Fig. 3 $\log \text{CGL} = 0.074 + 0.48 \cdot \log \text{BoW}$

of man (114) was close to this average. Thus, in simians including man, the HSE was on the average only slightly larger than in isoponderous prosimians. In contrast to this relative uniformity of prosimians and simians, there were several strong deviating species within these suborders. A HSE 2.1 times larger than in the average prosimian was found in *Tarsius* (SI = 213), and a HSE 1.4 times larger in *Loris* (SI = 140). The HSE of *Aotus* was about 1.8 times larger than in average prosimians and about 1.6 times larger than in average simians. All these species are night-active. The relatively largest HSE in a day-active simian was found in *Erythrocebus* (151) and the relatively smallest HSEs





Figs. 4—6. Size indices for half surface of the eye (HSE, Fig. 4), nervus opticus cross sectional area (NOA, Fig. 5), and corpus geniculatum laterale volume (CGL, Fig. 6). Reference bases are the thick lines given in Figures 1—3. Each of the small horizontal bars represents the size index for a species. On the left the range (vertical bars) and the mean (o) of various groups are given, while on the right, the sequences of means for the various groups are scaled.



in such extremely different forms as the Callitrichidae and *Pongo pygmaeus* (75—85). Within the prosimians the lowest HSEs were found in *Perodicticus potto* (73) and the Cheirogaleidae (74—86).

Distinctly smaller HSEs than in all the primates were found in the tree-shrews (Scandentia) and especially in Insectivora (Fig. 4). Compared with isoponderous average prosimians, tree-shrews have HSE indices of less than $^{1}/_{2}$, hedgehogs of about $^{1}/_{5}$, shrews of about $^{1}/_{60}$ and *Talpa* of about $^{1}/_{160}$ (Table 2, column 1). Based on the data of Table 1 (column 1), ratios between the HSE in *Talpa* and the HSEs in roughly isoponderous prosimians are 1:117 for *Microcebus* and 1:186 for *Galagoides*, thus confirming by direct comparison the coefficient ($^{1}/_{160}$) for the HSE indices.

Within the domestic and/or laboratory animals, the HSE indices of dogs and cats were about 1.3 times that of the average prosimians, i.e. in the upper range of those of simians. The HSE index of the rabbit was close to the prosimian average and those of the rat and mouse were distinctly lower (about $^1/_4$ and $^1/_9$ of the prosimian average).

Size indices of the optic nerves: Cross sectional areas of the optic nerves (NOA) are given in Table 1, column 4. Their allometric comparison provides additional information on the comparative development of the visual structures since: (1) the optic nerve is the only channel for visual output of the retina and (2) our collection of NOA data is larger and less heterogeneous than that of eye data.

When the NOA is plotted against body weight in a double logarithmic scale, the slope of the regression line of all prosimian species (n 18) is 0.283 and that of all simian species (n 29) is 0.317. The average is 0,30 and therefore the reference line was given this slope. A line with this slope was drawn through the geometric mean of body weights and NOAs of the prosimians (Fig. 2). For each body weight this line determines a nervus opticus area which represents the prosimian average.

The size indices related to this prosimian base line (A = 0.30; B = —0.611) are given in Table 2, column 2 and in Fig. 5. The average index of prosimians is 99 without *Tarsius* (n 18) and 111 with this genus. The lowest index was found for *Perodicticus potto* (46), the highest for *Tarsius* (316). The second maximum was found in *Lemur catta* (159). All the day-active prosimians had relatively high indices ranging from 112 in *Propithecus* and *Indri* to 159 in *Lemur catta*. The average was 134 (n 5). In the night-active prosimians, besides the extremely high value for *Tarsius* (316), high values were found in *Galago senegalensis* (143), *Galagoides demidoff* (124) and *Avahi laniger occidentalis* (113), while all the other species had indices lower than 100. The average of night-active prosimians was 85 without *Tarsius* (n 12) and 103 with this genus. Within the Lemuridae, the night-active *Lepilemur* had a distinctly lower index (59) than the day-active species (av. 149; n 3). Within the Lorisidae, the Galaginae had on the average a distinctly higher value (av. index 120; n 3) than the Lorisinae (av. 65; n 3).

The average index of simians was 352 (n 29), i.e. the NOA of simians was on the average 3.5 times larger than that of isoponderous prosimians. The highest index was found for *Papio papio* (570) and the lowest for the night-active *Aotus trivirgatus* (192). With the exception of *Tarsius* (index = 316) there was no overlap between prosimians and simians. Average values were: 296 for Callitrichidae (n 3), 287 for Cebidae with *Aotus* (n 6) and 306 without this species, 395 for Cercopithecidae (n 14), 343 for Pongidae (n 5) and 335 for man. Within Cercopithecidae, the average indices were different for Colobinae (283, n 2) and Cercopithecinae (414, n 12); within the latter, the highest indices were found for the baboons (av. 549, n 9).

When the NOA indices (Fig. 5) are compared with the HSE indices (Fig. 4), they are in most cases similar in prosimians but in some cases extremely divergent. Extreme differences were found in *Galago senegalensis* in which the HSE was very small (index = 84; calculated from the data of SCHULTZ, 1940), whereas NOA was relatively large (index = 143) and in *Loris tardigradus* in which the relation was reversed (indices 140 and 68). In simians, the NOA indices of all species except *Aotus* were distinctly higher. In *Aotus* the NOA index was only slightly higher than the HSE index (192 versus 179).

The average NOA index of tree-shrews was about 2.5 times larger (253; n 2), than the average of isoponderous prosimians, whereas that of Insectivora was only $^1/_{11}$ (9.1, n 16). Within the Insectivora, the average NOA indices were: 24 in hedgehogs (n 3), 9.3 in hedgehog-like tenrecs (n 4), 4.4 in shrews (n 6) and 2.9 in moles (n 3). Thus the NOA of hedgehogs is about $^1/_4$ that of isoponderous prosimians, that of tenrecs about $^1/_{11}$, that of shrews about $^1/_{23}$ and that of moles about $^1/_{35}$.

Within the domestic and/or laboratory animals, the NOAs of mice and rats (18 and 19) were roughly on the level of that of hedgehogs, those of rabbits (67), on the level of night-active prosimians and those of cats and dogs (153 and 180), slightly above the level of day-active prosimians but below the level of simians.

Ratios between half surface of eye and nervus opticus area: The ratio HSE/NOA ranged from 48:1 in *Papio* to 842:1 in *Loris* (Table 2, column 4, and Fig. 7). *Papio* has a relatively small eye or a large optic nerve, *Loris* a relatively large eye or a very small optic nerve. A ratio of 231 was found in *Talpa*. The average values were 237 for shrews (n 3), 306 for hedgehogs (n 3) and 266 for all 7 species of Insectivora. Tree-shews had 72 (n 2), prosimians 432 (n 13; ranging from 205 in *Lemur catta* to 842 in *Loris*), simians 107 (n 27; ranging from 48 in *Papio* to 357 in *Aotus*). Within prosimians, the night-active species (n 11) had distinctly higher ratios than the day-active (n 2) (470 versus 224). A similar difference was found in simians, where the only night-active species *(Aotus)* had the by far highest ratio (357 versus a range from 48 to 139). The maximum value for day-active simians (139) was still distinctly lower than the minimum in prosimians (205). The range of the ratios of Insectivora were at the level of prosimians,

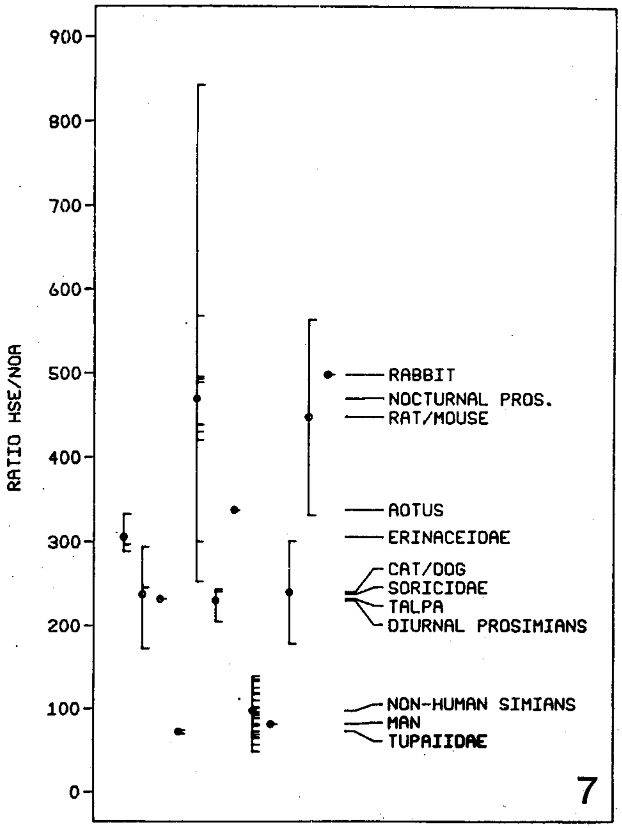


Fig. 7. Ratios between half surface of the eye (HSE) and nervus opticus cross sectional area (NOA). Each of the small horizontal bars represents the ratio for a species. On the left, the range (vertical bars) and the mean (o) of various groups are given, while on the right the sequences of means for the various groups are scaled.

those of tree-shrews at the level of simians. The HSE/NOA ratios will be discussed later in more detail (see Section 3 of Discussion).

Chiasma opticum

Mid-sagittal sectional areas of the chiasma opticum are given in Table 1, column 7. Regression analyses were performed and an average slope close to 0.26 was found. The size indices were similar to those of NOA and are, therefore, not given.

Ratios between chiasma opticum area and nervus opticus area: In Insectivora, the average ratios between the mid-sagittal sectional area of the optic chiasma and the cross-sectional areas of the optic nerves of both sides (CH II/NOA; Table 2, column 5) were 0.97 for Tenrecinae (n 4; range 0.87—1.09), 0.62 for Erinaceidae (n 3; range 0.58—0.68), and 1.17 for Soricidae and Talpidae (n 9; range 0.55—1.82). In Soricidae and Talpidae, both optic nerve and chiasma are very small and difficult to measure exactly. Thus the very wide range of the ratios may be at least in part due to these difficulties. The variation in Tenrecinae and Erinaceidae, where the structures are larger, was by far less. In Scandentia and Macroscelidea, ratios distinctly larger than 1 were found. The average was 1.68 for Scandentia (n 2) and 1.63 for Macroscelidea (n 2).

In prosimians, the ratios varied around 1 with an average of 1.02 (n 18). The range was from 0.63 in *Daubentonia* to 1.42 in *Cheirogaleus major*. There were no differences between night-active (n 13) and day-active species (n 5). The average values were 1.21 for Cheirogaleidae (n 3), 1.15 for Lemuridae (n 4), 0.79 for Indriidae (n 3), 1.17 for Lorisinae (n 3) and 0.97 for Galaginae (n 3). The *Tarsius* average was 0.71. Thus in *Daubentonia*, *Tarsius* and Indriidae, the chiasma is on the average smaller than both optic nerves together; in Lemuridae, Lorisinae and Cheirogaleidae it is larger. Whether the differences in the average ratios between prosimian families will be confirmed when larger samples are investigated is an open question. For the present, a discussion of these differences would be premature.

In New World monkeys, the ratios ranged from 0.77 to 1.04 with an average of 0.89 (n 9). Corresponding values for Old World monkeys were 0.55—1.17 (av. 0.90; n 10), and for apes 0.55—0.75 (av. 0.63; n 3). In man, the ratio was 0.38.

Tractus opticus

The optic tract volumes (TRO) are given in Table 1, column 9. They provide an additional measure of the size of the optic pathways, however, the values depend not only on the thickness of the bundle but also on its length.

Interstructural regression TRO versus CGL: There is a good correlation ($r^2 = 0.985$) between tractus opticus volume and corpus geniculatum laterale volume (n 35). Insectivora, prosimians, simians and man are all on a common regression line regardless of their different taxonomical groups and evolutionary levels. The slope was 1.02 indicating that from smaller to larger species the tractus opticus volume increases slightly more than the corpus geniculatum laterale volume. This may be due to a greater length of the bundle and somewhat larger thickness of fibers in enlarging brains. The greatest deviations from the regression line were found in *Tupaia glis* and in some night-active prosimians *(Loris*, *Perodicticus* and *Tarsius)*. In *Tupaia*, this deviation is due to a relatively large optic tract volume as plotted against body weight (data not given). In the three prosimian species mentioned above, the size indices of the optic tract were small or at least distinctly smaller than those of the lateral geniculate body, whereas in the other prosimian species, the indices of the tractus opticus were similar to or larger than those of the lateral geniculate body.

Corpus geniculatum laterale

Volumes of the dorsal nucleus of the corpus geniculatum laterale (CGL) are given in Table 1, column 10.

Size indices: When the corpus geniculatum laterale was plotted against body weight in a double logarithmic scale, the slope of the regression line of all prosimian species was 0.474 (n 18), and that of all simian species was 0.487 (n 21). The slope of the regression lines of the eight primate families under investigation (with more than one species) had an average of 0.470. The average of the given three values is close to 0.48 and therefore the reference line was given this slope. A line with this slope was drawn through the geometric mean of body weights and CGL volumes of the prosimians (Fig. 3). For each body weight this line determines a CGL volume representative of the prosimian average.

The size indices related to this prosimian base line (A = 0.48; B = 0.074) are given in Table 2, column 3, and in Fig. 6. The average index of prosimians was 100 without *Tarsius* and 104 with this genus. The lowest index was found for *Lepilemur* (50) and the highest for *Tarsius* (173). The second maximum was found in *Lemur albifrons* (135). The four day-active prosimians had relatively high indices, ranging from 109 in *Propithecus* to 135 in *Lemur albifrons* with an average of 122. In the night-active prosimians, besides the very high value in *Tarsius* (173), high values were found in *Loris* (130) and *Nycticebus* (122). The average of night-active prosimians was 93 without *Tarsius* and 99 with this genus. Within the Lemuridae, the night-active *Lepilemur* was extremely divergent from the day-active forms (50 versus 130).

The average index for simians was 189 (n 21), i.e. the CGL volume of simians was on the average 1.9 times larger than that of isoponderous prosimians. The highest indices were found for *Miopithecus* (307) and *Papio anubis* (258) and the lowest for the night-active *Aotus* and for *Alouatta* (110 each). The average values were: 156 for Callitrichidae (n 4), 183 for Cebidae (n 6), 221 for Cercopithecidae (n 7), 177 for Pongidae (n 3) and 172 for man. Within the Cercopithecidae, the index of *Colobus* (Subfam. Colobinae) was distinctly lower than the average of the Cercopithecinae (154 versus 232).

When the CGL indices (Fig. 6) were compared with the HSE indices (Fig. 4) they were generally similar in prosimians. In a few cases, the CGL indices were clearly larger *(Cheirogaleus major, Galago senegalensis)* or smaller *(Otolemur crassicaudatus, Tarsius)*. In all simians (except *Aotus*), the CGL indices were larger, and in most cases, distinctly larger. In contrast to this general simian line, the CGL index of *Aotus* was distinctly smaller than its HSE index (110 versus 176).

In relation to the prosimian base line, the CGL index of isoponderous *Tupaia* spp. (n 2) was about 100, i.e. of similar size, but that of *Urogale* was only somewhat more than half size (Fig. 6). The Insectivora have distinctly smaller CGLs than isoponderous prosimians. The average indices were 2.8 for Soricidae (n 5), 4.5 for Tenrecinae (n 4), 15.0 for Erinaceidae (n 2), and 5.6 for the total of the eleven Insectivora species. Thus the CGL of hedgehogs was about $^1/_7$ that of isoponderous prosimians, that of hedgehog-like tenrecs about $^1/_{22}$, that of shrews about $^1/_{36}$, and that of Insectivora about $^1/_{18}$.

Discussion

(1) Regression analyses

The range of systematic groups (families, suborders, orders) across which a calculation of a common slope is meaningful was discussed in Part I. The restrictions stressed there can be demonstrated, for example, in the plot CGL versus body weight (Fig. 3) where families and primate suborders lead to a slope close to 0.48, whereas for the total of primate species, a slope of 0.56 was found, and for Insectivora and Primates together, one of 0.88. When using a reference line with one of the slopes of the higher taxonomic units, the influence of body size is not minimized and within narrowly related groups, the smaller species would have larger indices than the larger species: existing differences in relative CGL size may be concealed. Therefore the slopes of all regression lines from narrowly related groups (families and even subfamilies) must be checked upwards through larger systematic groups (suborders, orders), and finally from the total of all mammalian species so far investigated. Decisions as to which category of systematic units to use to establish the slope of the reference line should be based on such a test. The position of the reference line can be established on utilitarian grounds, e.g. through the average of Insectivora, prosimians or others. When plotting structures which are more or less dependent on one another, all the regression lines and their slopes may be similar and a common regression line may be valid for all mammalian species (see interstructural regression TRO versus CGL).

When the visual structures are plotted against body weight, the slope of the regression lines within narrowly related groups are generally low and include the lowest slopes so far found in such plots. The average slopes are 0.20 for HSE, 0.26 for CH II, 0.30 for NOA, 0.48 for CGL and 0.52 for TRO. HSE, CH II and NOA are areal measures. By way of calculation, corresponding (partly theoretical) volume measures would have the following slopes: EYV 0.30, CH II 0.39 and NOA 0.45.

The slopes of the regression lines of the eyes (HSE, EYV) are especially low and indicate that with increasing body weight, the increase in eye size is relatively low. The data of FRAHM (1973), who compared eye weights of hamsters of different body weights *(Phodopus, Mesocricetus, Cricetus)*, result in a slope of 0.316, which is similar to our own (EYV 0.30). According to FRAHM, the eyes had by far the lowest slope found for various organs in hamsters.

Data on eye size of Insectivora and Primates are given by many authors (Table 1, column 2) but mainly by SCHULTZ (1940), ROCHON-DUVIGNEAUD (1943) and ROHEN (1962). Calculated from the data of SCHULTZ, male and female primates have similar eye size, despite the large sex differences in body weight in several species. The general rule is, however, that the larger the species, the larger the eyes. SCHULTZ expressed relative eye size by calculating eye volume per unit of body weight (e.g. mm^3 eye per g body, or cm^3 eye per kg body) and found values between 0.06 in male orang-utans and 22.43 in a female tarsier.

The relative values resulting from this type of comparison are generally low in large species and generally high in small species; thus they are highly influenced by differences in body size. SCHULTZ (1940) noted this influence but did not introduce methods to minimize it, e.g. allometric methods. If the type of comparison used by SCHULTZ is transformed into an allometric equation for a double logarithmic diagram[1], the reference base line would have a slope of 1. This, however, is by far too high as has been pointed out in preceding paragraphs, where slopes close to 0.30 were found for eye volumes in narrowly related groups.

(2) Comparative size of non-cortical visual structures

Insectivora: All the Insectivora have small visual structures as compared with those of primates. There are, however, distinct differences between the families so far investigated.

Talpidae: It is not surprising that the lowest indices, i.e. the relatively smallest visual structures, were found in the mainly subterranean European mole *(Talpa europaea)*. Based on statements of former investigators, moles have rudimentary eyes which, however, enable them to perceive moving objects, to distinguish day and night, and to cross rivers by visual guidance (ROCHON-DUVIGNEAUD, 1943; QUILLIAM, 1966). Responses to visual stimuli have been found by LUND and LUND (1965) but nearly none by KRISZAT (1940). Detailed descriptions of the histology of the mole's eye have been given by ROCHON-DUVIGNEAUD (1943), QUILLIAM (1966), SIEMEN (1976), and SATO (1977). According to the indices found for the optic nerves, the desmans have somewhat larger visual structures, similar to those of shrews.

Soricidae: Based on our allometric comparisons, shrews have small visual structures which, however, are distinctly larger than those of European moles (HSE indices 2–3 times, NOA indices 2–6 times). According to ROCHON-DUVIGNEAUD (1943), shrews have very small eyes and poor vision. The eyes are not rudimentary, but their tiny size may reduce the animal's visual acuity. GRÜNWALD (1969) found that vision plays a subordinate role in the orientation of *Crocidura*. Distinction of light and dark and directional vision were shown to be used in orientation, but no indications of a participation of form vision were found. GRÜN and SCHWAMMBERGER (1980) summarized behavioral studies on *shrews* and on the histology of their eyes. They found functioning retinae in *Sorex*, *Neomys* and *Crocidura* but a slight degeneration of receptor cells.

The HSE/NOA convergency of shrews is similar to that of day-active prosimians. This is in accordance with the periods of activity, which in shrews include daytime.

Erinaceidae: Of all the recent Insectivora so far investigated, hedgehogs have the largest visual structures. The HSEs are on the average about 30 times larger than in the European mole of (theoretical) equal body weight and 12 times larger than in shrews. The optic nerves are about 20 respective 5.6 times larger, and the CGLs of hedgehogs are on the average 5.5 times larger than those of isoponderous shrews. This is compatible with the statement of POLYAK (1957) that the vision of hedgehogs

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1 Relative size $RS = \frac{EYE}{BoW}$; $EYE = RS \cdot BoW$;

$$\log EYE = \log RS + \log BoW = \log RS + 1 \cdot \log BoW$$

is definitely improved over that of shrews. HERTER (1933, 1935) found in experimental behavioral studies that hedgehogs are able to differentiate shapes and colours, and according to HALL and DIAMOND (1968), pattern discrimination is present. SIEMEN (1976), however, questions the capacity for colour vision.

It is worth mentioning that in all three species of hedgehogs low CH II/NOA ratios were found; this may indicate a partial crossing of optic fibers in the chiasma. The HSE/NOA convergency of hedgehogs is similar to that of night-active primates. This is in accordance with the period of main activity, which in hedgehogs is concentrated at nightfall and night.

Tenrecinae: No eye measurements of the hedgehog-like tenrecs of Madagascar are available, but according to the indices of both optic nerves and CGL, the relative size of the visual structures is intermediate between that of shrews and of hedgehogs, although somewhat closer to the shrews (see Fig. 5 and 6). HERTER (1962) described some visual reactions of *Echinops telfairi* on the basis of behavioural investigations.

Scandentia: Following the "World List of Mammalian Species" of CORBET and HILL (1980), the tree-shrews are classified as a separate order with the single family Tupaiidae. Some of the size indices and ratios of the visual structures of tree-shrews show special features which are not found in both Insectivora and Primates and point to the separation of tree-shrews from these orders.

Tree shrews have distinctly larger visual structures than hedgehogs (also CAMPBELL et al., 1967). Compared with prosimians, their eyes are relatively small but their optic nerves very large. The HSE indices are less than half that of isoponderous average prosimians, whereas the NOA indices are 2 to 3 times larger. As a consequence, the HSE/NOA convergency is distinctly less than that of prosimians (about $^1/_6$). Since, however, the tree-shrews are purely day-active, which is in contrast to the night-activity of the majority of prosimian species, a comparison of their HSE/NOA ratios with those of day-active primates is more appropriate. In many higher simians, ratios like those in tree-shrews and even lower values are found (Table 2, column 4; Fig. 7). However, in most of these day-active simians, both the eyes and optic nerves are larger than in (theoretically) isoponderous tree-shrews. Whereas the indices of the optic nerves of tree-shrews reach slightly into the range of those of day-active simians, the eyes of tree-shrews are only half the size or less.

Another special feature of the visual structures of tree-shrews is the relatively high CH II/NOA ratio, which points to a high percentage of crossing fibers and is in accordance with statements of TIGGES (1966), CAMPBELL et al. (1967), LAEMLE (1968), and GIOLLI and TIGGES (1970).

The large relative size of the optic paths (pointed out also by CAMPBELL et al., 1967; LAEMLE, 1968; and BARON, 1981) is in contrast to the distinctly smaller relative size of the CGL: (1) We found for *Tupaia glis* the greatest deviation from the TRO versus CGL regression line (see above) since in this case TRO is very large and CGL very small, and (2) the differences between NOA and CGL indices are larger in tree-shrews than in primates. BARON (1981) described similar strong differences between TRO and CGL indices. These relationships agree with the results of TIGGES (1966) that only about 20% of the optic fibers terminate in the CGL, with the main part ending in the optic tectum.

There is good functional evidence of pattern discrimination and colour discrimination in *Tupaia glis* (TIGGES, 1961, 1963, 1964; DIAMOND, 1967; SHRIVER and NOBACK, 1967; LAEMLE, 1968).

Prosimian, Primates, Cheirogaleidae: The relative size of all the visual structures of this night-active family of Madagascar is lower than the prosimian average. The HSE/NOA convergency is high, which is in accordance with night-active behaviour.

Lemuridae: Within this family, the NOA and CGL indices of the day-active *Lemur* spp. and *Varecia* are more than two times larger than those of the night-active *Lepilemur*. Unfortunately, eye data from *Lepilemur* are not available. The NOA and CGL indices of *Lepilemur* are distinctly below the prosimian average, and the CGL index is the lowest found within primates. This is in general agreement with results about the whole brain of *Lepilemur* and many of its parts, which are relatively smallest within primates (STEPHAN, 1967, 1972). In contrast, the day-active *Lemur* and *Varecia* have optic nerves and CGLs which, within prosimians, are very large, whereas the eye size is close to the prosimian average. Their HSE/NOA convergency is the lowest within prosimians, which is in keeping with their day activity.

Indriidae: We have no eye data from this family. They would be of some interest since *Avahi* is night-active, whereas *Propithecus* and *Indri* are day-active. No differences were found in the relative size of the optic nerves, but the relative size of the CGL is somewhat larger in the day-active than in the night-active species. This corresponds with differences found for day- and night-active Lemuridae, but is less pronounced.

Daubentoniidae: Within prosimians, *Daubentonia* has the largest brain in relative terms and many of the brain parts are also largest (STEPHAN, 1967, 1972). Such an exceptional position is lacking in the visual structures, where the indices are only somewhat larger than the prosimian average for HSE and CGL, but somewhat smaller for NOA.

Lorisidae: The eyes of the two subfamilies are on the average of similar size, whereas the relative size of the optic nerves is nearly two times larger in the Galaginae than in the Lorisinae. Thus, the HSE/NOA convergency is distinctly higher in Lorisinae than in Galaginae.

Within the Lorisinae, *Loris* has relatively large eyes which are associated with some unique characteristics in skull and brain (SPATZ, 1970). In contrast, the optic nerves are very small and thus the HSE/NOA convergency is extremely high, in fact, the highest found so far. In *Perodicticus,* all the visual structures are relatively small and still smaller than in the Cheirogaleidae. SPATZ (1970) pointed out that the development of potto's visual structures is low within lorisids. Unusually small eyes were already noted by SCHULTZ (1940) who suggested that *Perodicticus* may rely more on its senses of touch and smell than on vision for its comparatively sluggish nocturnal activities.

Within the Galaginae, the very low HSE/NOA convergency of *Galago senegalensis* is due as much to the relatively large nervus opticus as to the relatively small eye. However, the number of specimens so far investigated is low and from different sources and thus may not be well founded.

Tarsiidae: Tarsius is a highly specialized visual animal (POLYAK, 1957, p. 933) and all its visual structures are large. According to KOLMER (1930) and WALLS (1953), *Tarsius* has the largest eyes relative to body size of any mammal. Our results confirm this at least for the mammalian species investigated here. Each of the eyes has approximately the same volume as the total brain. The optic nerve appears to be relatively small when seen together with the eyeball. Despite this impression, measurements made by STEPHAN (1969) and BARON (1979, 1981) showed large size indices for chiasma and optic tract which are now confirmed by the measurements on optic nerves. The results are in accordance with statements made by WOOLLARD (1925) and HILL (1955) on the large size of the optic nerve. Compared with other prosimians, both NOA and CGL indices are largest in *Tarsius,* but when compared with simians, they are not. For NOA and CGL, the exceptional size as found for the eyes is lacking and both indices are within the range for simians and slightly below the simian average.

In *Tarsius,* the NOA index is distinctly higher than the HSE index (316 versus 213) and thus the HSE/NOA convergency is surprisingly low for a species which is highly adapted to night-vision.

Tarsius is the only prosimian species whose NOA index reaches into the range of the diurnal simians (Fig. 5). Another exception was described by ROHEN and CASTENHOLZ (1967), who found that the "centralization" (ROHEN, 1958) of the retina is lower in nocturnal than in diurnal primates, except for *Tarsius* whose eye has a well developed fovea centralis, though the retina is highly adapted to night-vision by an absence of cones and a multitude of rods with extremely long outer segments. Details on the histology of the *Tarsius* retina have been given by WOOLLARD (1925, 1926), KOLMER (1930), POLYAK (1957), ROHEN (1966), ROHEN and CASTENHOLZ (1967), and CASTENHOLZ (1984).

Simian Primates: Within the simians, the visual structures appear to be more uniform than within the prosimians with relatively few exceptions requiring discussion.

The largest deviations are found in *Aotus trivirgatus,* which is the only night-active species within simians and which occupies an extreme position in the relative size of all visual structures. The size of its eyes is by far the largest; the size of the optic nerves by far the smallest; and in the size of the CGL, *Aotus* shares the lowest place with *Alouatta.* Due to its large eyes and small optic nerves, the HSE/NOA convergency of *Aotus* is by far the highest within simians. This high value is typical for night-active species and, together with the very large eyes of *Aotus,* which are second to *Tarsius,* is evidence of the high adaptation of *Aotus* to night-activity.

Within the day-active simians, some species and/or groups have in one or more of the visual structures distinctly smaller relative size than is found for the average. Such species and/or groups are: (1) the marmosets, (2) some of the leaf-eating species, and (3) the great apes.

(1) In all marmosets (Callitrichidae), the relative eye size is at the lower limit, and both NOA and CGL size are in the lower half of the simian range giving a general impression that the visual structures of the marmosets are relatively poorly developed within the simian primates.

(2) The relative size of all the visual structures investigated so far is comparatively low in *Alouatta, Colobus* and *Nasalis.* This is in accordance with

results for other brain structures and the total brain (Stephan, 1967, 1972), which indicate that the brain development of *Alouatta* and *Colobus* is generally low. These are leaf-eating species and there seems to be a relationship between low brain development and leaf-diet (Bauchot and Stephan, 1969; Clutton-Brock and Harvey, 1980). However, in *Presbytis* spp., which are also members of the leaf-eating Colobinae, the eyes were found to be of typical or even large size within simians.

(3) In the great apes and especially in the orang-utan, the eyes are relatively small. Small NOAs are found in gorilla and orang-utan and a small CGL, in gorilla (not measured in orang-utan). The chimpanzee has higher NOA and CGL indices, similar to those of man. The reason for the low relative size of the visual structures may be similar to that of the leaf-eating monkeys, since the diet in orang-utan and gorilla is purely or at least predominantly vegetarian and includes leaves. Their movements both in trees and on the ground are leisurely and deliberate whereas gibbons and chimpanzees are more agile and sometimes leap in trees (Walker, 1964). These differences may also be related to the development of the visual structures.

Apart from the species discussed above and from the doubtful eye data of a single specimen of *Saimiri oerstedi* (given by Schultz, 1940), the visual structures of the day-active simians are larger than those of the average prosimians. Compared with average prosimians, the typical day-active simian (incl. man) has HSEs about 1.2 times larger, NOAs about 3.9 times larger, and CGLs about 2.2 times larger. There are no noteworthy differences in the indices between New- and Old-World species (man included). However, the HSE/NOA convergency, which is on the average 96:1, appears to be somewhat higher in New-World (120:1) than in Old-World simians (89:1). Whether such differences are real cannot be substantiated at present.

It may still be worth mentioning that in *Erythrocebus* especially large eyes were found and in the baboons especially large optic nerves. Both forms inhabit open regions and savannas, prefer to live on the ground and may have similar visual behaviour and capacities, in which orientation and detecting enemies over long distances may play a major role.

(3) *Size differences of visual structures in diurnal and nocturnal species*

Besides the clear differences in the HSE/NOA convergency between diurnal and nocturnal species discussed below, Baron (1981) found higher indices for the optic tract of diurnal species than in nocturnal primates, and Schulz (1967) found differences in the composition of the CGL. According to Schulz, the nocturnal primates have a distinctly higher percentage of the magnocellular components of CGL than the diurnal species, both in volume of the layers and in their cell numbers. Schulz supposed that the large celled components were related to scotopic vision. Furthermore, Hassler (1965, 1966), and Hassler and Wagner (1965) gave some functional interpretations and pointed out that both input and output of the magnocellular layers have thicker fibers and higher conduction speed. They suggested that the magnocellular layers are mainly related to the scotopic receptors and to the periphery of the retina. Due to the greater conduction velocity, excitations from the magnocellular layer reach the area striata earlier and produce action potentials earlier than do excitations from the parvocellular layers of the geniculate, probably serving to facilitate the adjustment of eye movements (Hassler, 1965). Investigations by Kaplan and Shapley (1982) have shown that neurones of the magnocellular layer have much higher contrast sensitivities than do the neurones of the parvocellular layer which, in turn, are involved in the analysis of coloured patterns. Rohen and Castenholz (1967) found that in general the "centralization" (Rohen, 1958) of the retina is lower in nocturnal primates than in diurnal (except *Tarsius*, see above).

(4) Convergency from eye to optic nerve

Half the outer surface of the eye (HSE) may give a rough comparative measure of the extent of the retina and the cross-sectional area of the nervus opticus (NOA) may provide a measure of the visual output of the retina. The ratio between these two parameters (HSE/NOA) may indicate the amount of convergency from the retina to the visual centers of the brain. The range of this ratio is from 48 : 1 in *Papio papio* to 842 : 1 in *Loris tardigradus* (Fig. 7).

In primates this ratio has a close relationship to the period of main activities. In the night-active species (11 prosimians, one simian) the ratio is on the average 459; in the day-active species (2 prosimians, 26 simians) it is 106. Thus, HSE per unit NOA is on the average 4.3 times larger in night-active species than in day-active species, with no overlap.

Within the night-active primates most species have ratios between 420 and 569. Only *Loris tardigradus* has a clearly higher ratio (842), whereas *Aotus trivirgatus* (338), *Tarsius* (300) and *Galago senegalensis* (252) have lower ratios. Since the eye size of Galago was calculated from data of one specimen only (given by SCHULTZ, 1940), it may not be well grounded. For *Galagoides demidoff*, which is distinctly smaller, ROHEN (1962) found larger eyes (see Table 1, column 1). *Loris* has very small optic nerves relative to its large eyes and thus seems to have the highest convergency from its retina to brain centers. *Tarsius* and *Aotus*, which have the largest eyes in relative terms (see Table 2, column 1), have relatively thick optic nerves for night-active species. In the case of *Aotus*, its affiliation to simians may play a role.

Similarly, in the group of the day-active primates, the prosimian species (*Lemur catta* and *Varecia variegata*) have distinctly higher ratios than the simian species (averages are 224 versus 97) without, however, reaching the lower limit of the range of the night-active prosimians. According to PARIENTE (1979), *Varecia* (ratio 243) is a purely diurnal species. Thus, in *Varecia*, the factor determining its intermediate position in the HSE/NOA ratios is likely to be its systematic affiliation or other considerations, rather than its crepuscular vision. The simian species (n 26) have a relatively narrow range of ratios going from 48 in *Papio papio* to 139 in *Callithrix jacchus*. Next to *Papio papio*, the second lowest ratio is found in *Papio anubis* and thus baboons generally appear to have low ratios. Since baboon eyes are of average size (for primates), the low ratios are due to very large optic nerves (compare columns 1 and 2 of Table 2).

Low ratios were also found in the tree-shrews (av. 72; n 2) in which relatively small eyes are connected with the brain by relatively thick optic nerves. Domestic and laboratory animals have ratios similar to those of prosimians. The ratios of cats, mice, rats and rabbits are similar to those of the night-active group, while the ratio of the dogs is somewhat lower than those of the day-active prosimians.

Our findings are consistent with morphofunctional aspects of the retina. The retinas of animals with scotopic vision have larger receptive fields and therefore less ganglion cells (per unit area) sending their axons to the optic nerve than the retinas of diurnal animals. In fact retinas of diurnal animals which are characterized by high visual acuity require a greater amount of optic fibers projecting to the CNS.

(5) *Nervus opticus size and fiber number*

Is it possible to draw conclusions from the cross sectional area of the optic nerve to the number of fibers running in it? Or in other words: can intraindividual (and intraspecific?) interrelations of the two parameters be generalized for interspecific comparisons?

The size of the optic nerve is not only a function of the number of fibers but also of their size (thickness) and of the amount of non-nervous tissue. A generalization would be possible if (1) the relative amount of non-nervous tissues as well as the size of fibers were relatively constant in all the species, or (2) existing differences could be related to eye or brain or body size or other well-known parameters.

According to STONE and CAMPION (1978), the amount of capsule tissue in the cat is on the average 5.4% of the cross sectional area, the glial processes and blood vessels 20.0%, and thus the non-nervous tissues together 25.4% (4 nerves). In man, OPPEL (1963) found 19.0%, but did not specify whether the outer capsule of the optic nerve was included. So far the data given point to 19–25% non-nervous tissue and we assume that this portion is similar in various species.

The calibre of fibers within a given optic nerve vary considerably according to various investigations of several species. As a standard calibre mostly "modal diameters" of the fibers[1] are given. They are (in μm): 1.25 for opossum (HOKOC and OSWALDO-CRUZ, 1978); 1.0 for brush-tailed possum (FREEMAN and WATSON, 1978); 1.2 (OGDEN and MILLER, 1966), or 0.5 (POTTS et al., 1972) for rhesus monkey; 0.5 for man (POTTS et al., 1972); 2 (CHANG and CHENG, 1961), or 2–3 (VAN CREVEL and VERHAART, 1963), or 1 (HUGHES and WÄSSLE, 1976), or 1.1 (FUENTES et al., 1979) for cat; 0.9 for rat (FORRESTER and PETERS, 1967; DE JUAN et al., 1978); and 0.7–0.9 for mouse (HONJIN et al., 1977).

Different investigators arrive at different modal fiber calibres even for the same species (rhesus, cat). However, aside from the very low diameters given by POTTS et al. (1972) and the very high value for cats given by CHANG and CHENG (1961) and VAN CREVEL and VERHAART (1963), the modal diameters given are similar for all species and range from 0.8 μm in the mouse to 1.25 μm in the opossum. Because of the presence of thick fibers, the mean fiber diameter is larger than the modal diameter. In the brush-tailed possum, FREEMAN and WATSON (1978) found 1.6 versus 1.0 μm.

Assuming that the average fiber diameter is 1.6 μm and the non-nervous tissue is 25% of the cross sectional area of the optic nerve (NOA), we estimated

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1 The axon diameters are distinctly smaller. They are in percent of the fiber diameters: 60–87% in the brush-tailed possum (FREEMAN and WATSON, 1978), 60–80% in the rhesus monkey (OGDEN and MILLER, 1966), 50–72% in the cat (VAN CREVEL and VERHAART, 1963), and on the average 78% in the rat (FORRESTER and PETERS, 1967). Fibers with smaller diameters in general have lower percentages, i.e. the relatively thicker myelin sheaths.

fiber numbers for those species from which both NOAs and fiber numbers were given by previous investigators. The results of our calculations are only in a few cases close to the counts. In general they are distinctly higher, but in some cases clearly lower. One of the reasons for the generally high estimations may be that the disproportionally large cross-sectional areas of thick and very thick fibers could not be adequately taken into account.

General interrelations between the cross-sectional area of the optic nerve and the number of fibers are missed also when calculating fiber numbers per mm^2 NOA. The data used here were taken from the few investigations in which both parameters were given. The fiber numbers per mm^2 NOA vary from about 8,200 in dogs (Arey et al., 1942) to about 680,000 in mice (Honjin et al., 1977). However, even for the same species, different fiber numbers per unit were found. Thus in the cat, Hughes and Wässle (1976) found about 88,500 fibers per mm^2, but Stone and Campion (1978) estimated only about 59,000. Thus, differences in counts appear to be mainly due to differences in the methods.

Based on the existing investigations, it appears impossible to arrive at a well founded evaluation of the numbers of optic fibers by way of cross-sectional areas of the optic nerve. Nevertheless such relationships may exist and could be revealed by careful comparative investigation.

(6) Chiasma opticum size in relation to crossing fibers

According to Giolli and Tigges (1970), *Tupaia* has in relation with its ocular lateralization almost totally crossed optic nerves, whereas in primates, semidecussation of the nerves appear to have evolved with the establishment of ocular frontality. In primates, the amount of crossing fibers is from 50 to 60%. In contrast to the relatively constant percentages stated by Giolli and Tigges, we found a decrease of the CH II/NOA ratios from the lower to the higher primates. The average ratios show a general trend to become smaller from low Insectivora (average 1.05), through prosimians (1.02), monkeys (0.90), apes (0.63), and man (0.38). However, within all the groups the ratios show large variation and overlap. The differences were not found to be statistically significant and may disappear as larger and more homogeneous materials are investigated. If, however, the general trend really exists and the percentage of crossing fibers is more or less stable, the changes in the ratios must be caused by other factors such as (1) differences in the direction of the crossing fibers and/or (2) differences in the calibre of the crossing fibers, which in higher forms must be relatively smaller in relation to the calibre of the uncrossed fibers. To our knowledge there are no comparative investigations on this subject, and the question of whether the decrease in the CH II/NOA ratios is related to a decrease in the number of crossing fibers must remain open.

The very high ratios in Scandentia and Macroscelidea, and more generally, all ratios larger than 1, may indicate that the percentage of crossing fibers is close to 100 and/or the fibers may cross the midline more or less diagonally.

The surprising finding of similar ratios in hedgehogs and apes is in clear contrast to other Insectivora; and this raises the question of whether a good deal of the optic fibers in hedgehogs cross the midline in the chiasma, or whether other factors may cause the low CH II/NOA ratios.

(7) Phylogenetic aspects

Hedgehogs are of special interest with regard to primate evolution since "primitive insectivores of more or less erinaceoid stamp gave rise to the primates" (Simpson, 1945, p. 177). According to Polyak (1957, p. 869), the small eyes and poor vision of modern Insectivora implies that similar features may have been the original condition of the incipient mammalian phylum. In this context, a comparison of the visual structures of hedgehogs with those of primates is of interest.

Based on our data, the HSEs of prosimians are on the average about 5.4 times larger than those of isoponderous hedgehogs and those of simians about 7.1 times larger. The corresponding data for the optic nerves are 4.6 and 14.4, and for the lateral geniculate bodies, 6.9 and 12.6. These data indicate a clear enlargement of the visual structures from lower Insectivora (hedgehogs) through prosimians and simians and are compatible with the general conviction that in the phylogeny of primates, vision has become the dominant sensory system. Within the simians, however, there is no more general enlargement of the visual structures. Instead, the values of apes and of man are lower than those of the simian average.

According to the size differences between day- and night-active species, night-activity makes high demands on the size of the retinal surface (and thus eye size), whereas in day-active species, optic nerves and CGLs are distinctly more progressive. Thus, the size of the various non-cortical visual structures depends mainly on functional requirements, and there is no evidence for an enlargement related to increasing evolutionary levels.

Addendum

Valuable data on eye size in shrews are given by BRANIŠ (Acta Univ. carolinae — Biol. 1979, 409—445, 1981). Since they came to late to our knowledge, HSE values and indices based on these data are not included in Tables 1 and 2. Body weights (given by BRANIŠ), HSE values and indices are as follows:

	BoW	HSE	Index
0713 *Sorex alpinus*	10.9 g /	1.6 mm² /	0.6
0715 *Sorex araneus*	11.3 g /	4.1 mm² /	1.4
0769 *Sorex minutus*	4.9 g /	2.7 mm² /	1.1
0837 *Neomys anomalus*	13.6 g /	5.2 mm² /	1.7
0839 *Neomys fodiens*	17.2 g /	7.0 mm² /	2.2
0995 *Crocidura leucodon*	12.5 g /	4.5 mm² /	1.5
1085 *Crocidura suaveolens*	6.8 g /	4.1 mm² /	1.6

In spite of some larger deviations (e. g. in *Sorex minutus*) the eye diameters given by BRANIŠ result in indices which in range and average are similar to those given in Table 2 and discussed in the text. Relatively small eyes were found in *Sorex alpinus*, relatively large eyes in *Neomys fodiens*.

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The Parietal Cortex of Monkey and Man

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Im Mittelpunkt der vorliegenden Monographie steht die Funktion des Lobus parietalis des Cortex cerebri bei Primaten, im besonderen des Rhesusaffen (*Macaca speciosa*) und des Menschen. Die wesentlichen Abschnitte beschäftigen sich mit den Assoziationsfeldern im hinteren Teil des Parietallappens und schließen ein die Diskussion über die Ergebnisse des Autors am primären somatosensorischen Cortex. In den einzelnen Kapiteln werden Ergebnisse zur Entwicklungsanatomie, nach experimentellen Läsionen bei Versuchstieren und nach klinisch relevanten Läsionen sowie elektrophysiologischen Daten beschrieben und hinsichtlich ihrer funktionellen Bedeutung diskutiert.

Nach einer Einführung werden im II. Kapitel über ,,Anatomy and Evolution of the Parietal Lobe in Monkeys and Man" die Unterteilung des Lobus parietalis und die aktuelle Nomenklatur beschrieben. Der evolutionäre Aspekt der menschlichen Hirnentwicklung zeigt sich eindrucksvoll in der Vergrößerung des Parietallappens vom Rhesusaffen über den Schimpansen, über die frühen Formen der Homoniden bis hin zum rezenten Menschen als ein Ergebnis des Gebrauchs von Hand und Arm zum einfachen Nahrungsergreifen bis zum gezielten Gebrauch von Werkzeugen.

Kapitel III beschäftigt sich mit den funktionellen Eigenschaften der Neurone im primären somatosensorischen Cortex (S. I.), der seine Hauptprojektion vom ventrobasalen Thalamuskomplex erhält. Vordere und hintere Teile des S. I. sind mit den Hautrezeptoren bzw. den Rezeptoren der Tiefensensibilität verbunden. MOUNTCASTLE demonstrierte erstmals die columnare Organisation dieser Felder, die später von HUBEL und WIESEL auch für den visuellen Cortex dargestellt wurde. Mittels Mikroelektrodentechnik wurden unter verschiedenen experimentellen Bedingungen an nichtnarkotisierten Makaken die funktionellen Eigenschaften und Muster der Neurone untersucht. Diese zeigen eine funktionelle Differenzierung in antero-posteriorer Richtung des S. I., d. h. rezeptive Feldgröße und Komplexität der funktionellen Eigenschaften der Neurone nehmen in posteriorer Richtung zu (von Area 3 zu den Areae 1 und 2). Weitere Experimente belegen den Zusammenhang von bestimmten Verhaltensreaktionen und neuronalen Antworten in den Laminae des S. I.

Kapitel IV enthält eine Übersichtsdarstellung der wichtigsten bekannten Verbindungen des hinteren Teils des Parietallappens, die wesentlich von denen des vorderen Teils abweichen und diese Region als einen assoziativen Cortex mit Verbindungen zu über 60 corticalen Areae oder subcorticalen Kernen kennzeichnen. Diese Verbindungen gestatten nach Ansicht des Autors eine funktionelle Interpretation in dem Sinne, daß der hintere Parietallappen eine Rolle für Aufmerksamkeit und emotionelle Aspekte des Verhaltens sowie